Xinghao Peng

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Education

The Pennsylvania State University, University Park, PA 2024–Present
Ph.D. student, Industrial Engineering GPA: 3.94

Shanghai Jiao Tong University, Shanghai, China 2021–2024
M.Eng., Industrial Engineering, School of Mechanical Engineering

Shanghai Jiao Tong University, Shanghai, China 2017–2021
B.Eng., Industrial Engineering, School of Mechanical Engineering

Publications

- Peng, X., Liu, R., and Yang, H. "STEM: STructural EMbedding for Graph Representation of 3D Designs." *Manufacturing Letters 49 (2026): 1483–1493. doi:10.1016/j.mfglet.2026.08.054.*
- Peng, X., Liu, R., and Yang, H. "SCOUT: Spatiotemporal Coverage for Optimal Unmanned Tasking." *IEEE CASE 2025, pp. 2103–2108. doi:10.1109/CASE58245.2025.11163795.*
- Peng, X., Li, Y., and Tsung, F. "A graph attention network with spatio-temporal wind propagation graph for wind power ramp events prediction." *Renewable Energy 236 (2024): 121280. doi:10.1016/j.renene.2024.121280.*
- Peng, X. and Li, Y. "A wind power scenario generation method and its application based on spatiotemporal covariance functions." *Journal of Shanghai Jiao Tong University, 2023. doi:10.16183/j.cnki.jsjtu.2022.180.*
- Li, Y., Peng, X., and Zhang, Y. "Forecasting methods for wind power scenarios of multiple wind farms based on spatio-temporal dependency structure." *Renewable Energy 201 (2022): 950–960. doi:10.1016/j.renene.2022.11.002.*
- "Parsing Data Genome for Manufacturing Process Monitoring." *IISE Transactions (under review).*
- "Recurrence Analysis of Shock-Induced Operational Dynamics on Stochastic Networks." *International Journal of Production Research (under review).*

Research Experience

Doctoral Researcher, The Pennsylvania State University 2024–Present
Advisor: Prof. Hui Yang. Research in drone swarm control, cardiac simulation, and additive manufacturing.

Graduate Researcher, Shanghai Jiao Tong University 2021–2024
Advisor: Prof. Yanting Li. Studied economic dispatch for wind power grid integration using ramp-event prediction and scenario generation.

Visiting Researcher, Hong Kong University of Science and Technology Summer 2023
Advisor: Prof. Fugee Tsung. Researched wind power forecasting for intelligent wind farm management.

Undergraduate Researcher, Shanghai Jiao Tong University 2021
Advisor: Prof. Wei Qin. Studied integrated berth and quay-crane allocation for the Yangshan Port container terminal.

Teaching Experience

- Teaching Assistant, IE 522: Discrete Event Simulation, Penn State, Spring 2026.
- Teaching Assistant, IE 323: Statistical Models for Industrial Engineering, Penn State, Spring 2026.
- Teaching Assistant, IE 425: Stochastic Models in Operations Research, Penn State, Fall 2025 and Fall 2026.
- Teaching Assistant, Quality Management and Decision Theory and Methods, Shanghai Jiao Tong University, 2022–2023.

Honors and Awards

- Third Place, IEEE SMC 2026 AICA Competition.
- University Graduate Scholarship, Penn State, 2024.
- Outstanding Graduate Award (graduate), Shanghai Jiao Tong University, 2024.
- UAES Scholarship (top 20%), Shanghai Jiao Tong University, 2023.
- First-Class Academic Scholarship for Graduate Students (top 30%), Shanghai Jiao Tong University, 2022.
- Outstanding Graduate Award (undergraduate), Shanghai Jiao Tong University, 2021.
- Second Prize, National Industrial Engineering Application Case Competition, 2020.
- Second Prize, National College Students Energy Conservation and Emission Reduction Competition, 2019.

Industry Experience and Skills

Research Intern, Shanghai Pudong Development Bank Co., Ltd. Jan.–Apr. 2023
Developed federated learning models for metadata and researched causal inference in banking systems.

Operation Management Researcher, Dongfang Electric Wind Power Co., Ltd. Aug.–Sep. 2019

Computing: Python, MATLAB, C++, R, Minitab, Gurobi.

Languages: Chinese (native), English (fluent).